Module 8

Risk, Return, and the CAPM

FINE 3010, Financial Management

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Goals

- **Module 7** left two questions open,
  - What is risk?
  - How is the opportunity cost of an investment (which we use as the discount rate) related to risk?

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- This module answers both,
  - **Part 1**, returns and risk premiums
  - **Part 2**, return volatility and portfolio diversification
  - **Part 3**, beta, the risk that actually matters
  - **Part 4**, CAPM, the relationship between beta and expected return
  - **Part 5**, from expected return to discount rate

Module 8, Part 1

Returns and Risk Premiums

Return

Recall from **Module 4**

- **Return**, percentage growth of an asset's value over some period of time

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$$\text{Return} = \frac{\$180 - \$100}{\$100} = 80\%$$

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- **Expected return** $E[R_i]$, what the stock returns on average across the possible cases, where likelier cases count more
- To compute it, multiply each possible return by its probability, then add up,

$$E[R_i] = \text{Probability}_1 \times \text{Return}_1 + \text{Probability}_2 \times \text{Return}_2 + \cdots$$

Example, Expected Returns

- You want the expected return on a stock over the next year. Three things might happen,
  - A recession, 20% chance, the stock returns -10%
  - A normal year, 50% chance, the stock returns 5%
  - A boom, 30% chance, the stock returns 15%

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- Solution, the expected return is

$$E[R_i] = 0.2 \times (-10\%) + 0.5 \times 5\% + 0.3 \times 15\% = 5\%$$

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Its stock portfolio on June 30, 2026 ($20.2 billion),

Company	Business	Weight
Sandisk	Memory chips	28%
Micron	Memory chips	28%
Bloom Energy	Fuel cells	9%
TSMC	Chip maker	6%
Nebius	AI cloud	6%
CoreWeave	AI cloud	4%
Other holdings		19%
Total		100%

Source, SEC Form 13F, filed August 14, 2026

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$$R_P = w_1 \times R_1 + w_2 \times R_2 + \cdots + w_S \times R_S$$

- The expected portfolio return is likewise a weighted average of the securities' expected returns,

$$E[R_P] = w_1 \times E[R_1] + w_2 \times E[R_2] + \cdots + w_S \times E[R_S]$$

Example, Portfolio Returns

- One year ago, you bought two stocks for \$500 each, \$1,000 in total
- One stock returned 30% during the year, the other -10%
- What is your portfolio return over the year?

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- Solution, you invested $500/1{,}000 = 50\%$ in each stock, so the weights are 50% each,

$$R_P = w_1 \times R_1 + w_2 \times R_2 = 0.5 \times 30\% + 0.5 \times (-10\%) = 10\%$$

Example, Expected Portfolio Returns

- You want the expected return on your portfolio over the next year. You hold two stocks, 50% of your money in each
- The first stock's expected return is 5%, the second's is 9%
- What is your portfolio's expected return over the next year?

Example, Expected Portfolio Returns

- You want the expected return on your portfolio over the next year. You hold two stocks, 50% of your money in each
- The first stock's expected return is 5%, the second's is 9%
- What is your portfolio's expected return over the next year?

- Solution, the expected portfolio return is

$$E[R_P] = w_1 \times E[R_1] + w_2 \times E[R_2] = 0.5 \times 5\% + 0.5 \times 9\% = 7\%$$

The Risk-Free Rate and Market Returns

- $r_{f,t}$, the **risk-free rate** as of time t
  - The return from an investment with no risk, i.e., return is the same in every scenario
  - In practice, the interest rate on Treasury bills (short-term U.S. government debt)

The Risk-Free Rate and Market Returns

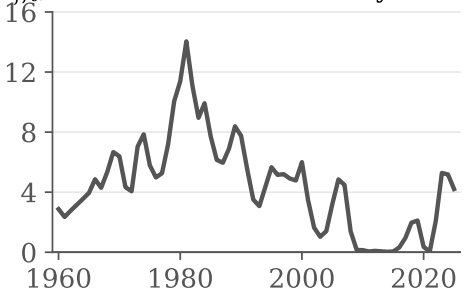
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- $R_{\text{mkt},t}$, the return on the market portfolio in period t
  - **Market portfolio**, the (theoretical) portfolio of all investable assets in the economy, weighted by market capitalization
  - In practice, represented by a broad market index such as the S&P 500

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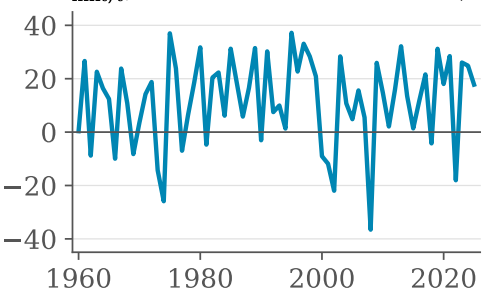
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- $E[R_{\text{mkt}}]$, the **expected return** on the market portfolio

The Risk-Free Rate and Market Returns over Time

$r_{f,t}$, interest rate on Treasury bills (%)

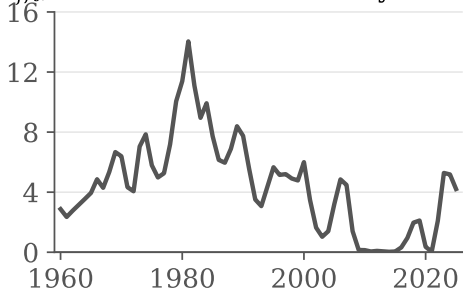


$R_{\text{mkt},t}$, return on the S&P 500 (%)

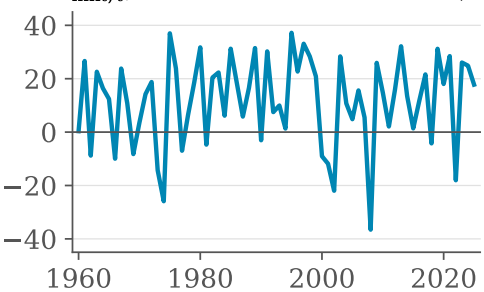


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- $r_{f,t}$ and $R_{\text{mkt},t}$ can be different at different t

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Treasury bills	3.4%	0%

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- Stocks are riskier than bonds, which are riskier than Treasury bills
- Naturally, we expect riskier assets to have higher expected returns

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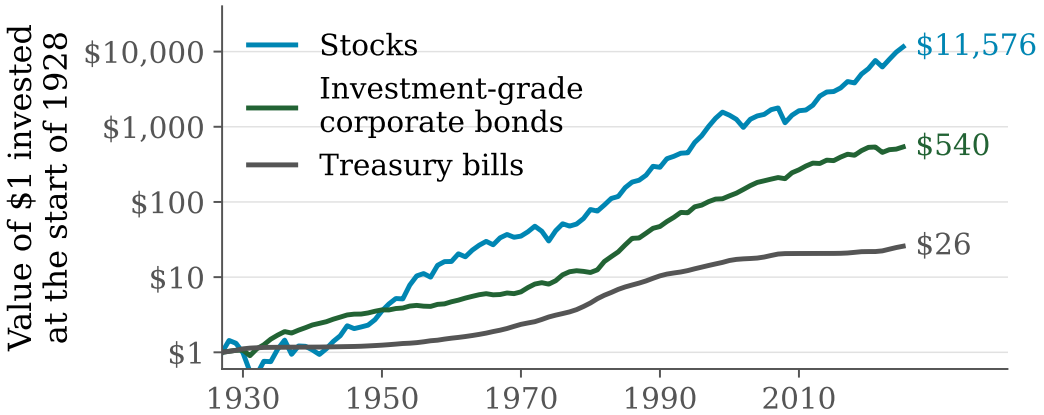
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$$E[R_i] = r_f + \text{Risk premium}_i$$

- The average extra return over Treasury bills is our estimate of the risk premium
- For investment-grade corporate bonds that is 3.5%, for stocks 8.5%

Returns over Time for Stocks, Bonds, and Treasury Bills



Recap of Part 1

- **Return**, percentage growth of an asset's value over some period of time

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- **Portfolio return**, the weighted average of the securities' returns
- $E[R_i] = r_f + \text{Risk premium}_i$, riskier assets have higher expected returns
- The risk premium of stocks is about 8.5%

Module 8, Part 2

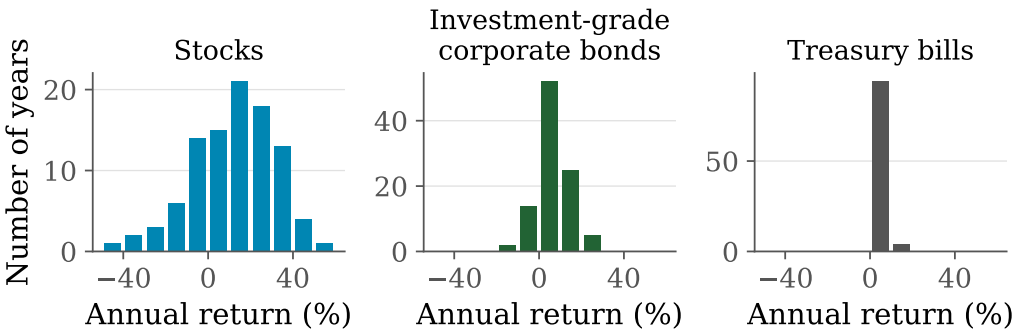
Return Volatility and Portfolio Diversification

The Risk Premium of Stocks Is High, but What about Volatility?

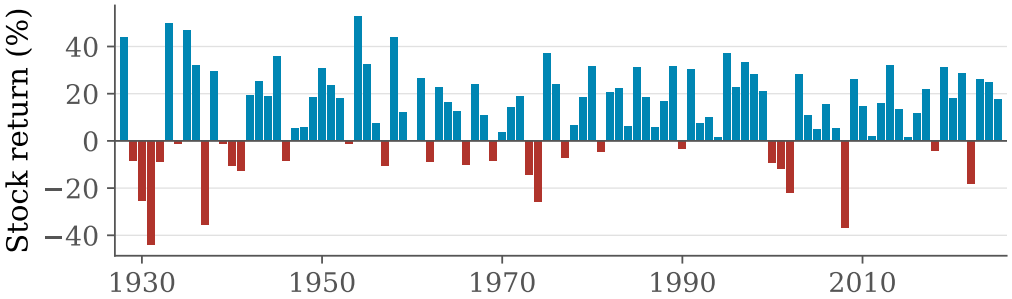
- How volatile are portfolios of stocks vs. bonds vs. Treasury bills?

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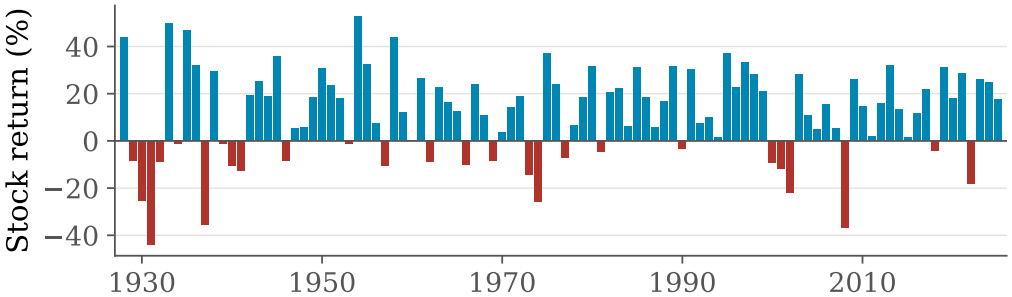
- How volatile are portfolios of stocks vs. bonds vs. Treasury bills?
- Consider the variation in annual returns, 1928–2025, how many years fell in each range,



Stock Market Returns over Time

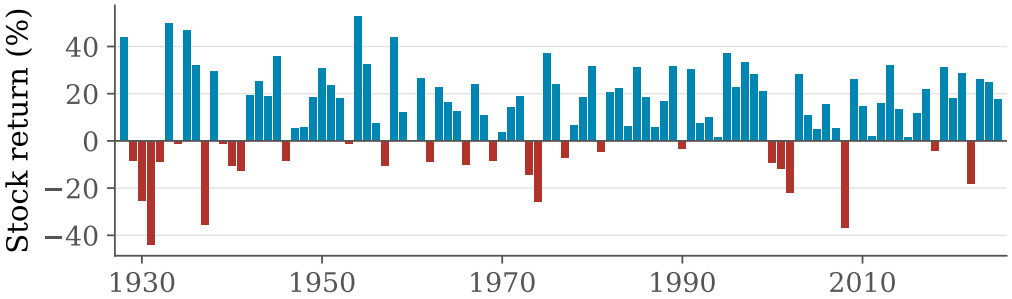


Stock Market Returns over Time



- A portfolio of all stocks has had negative returns in around 25% of all years

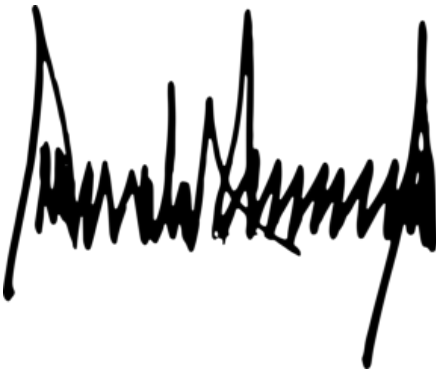
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- Individual stocks are even more volatile, around 50% of stock-years have negative returns

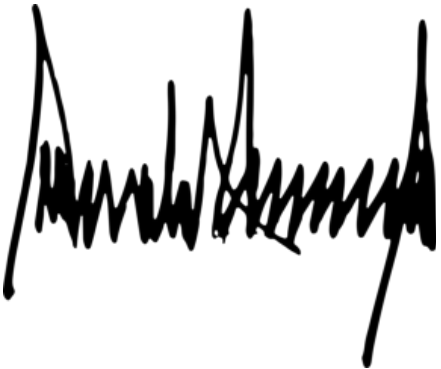
Discuss, Do They Look Alike?

US President Donald Trump's Signature

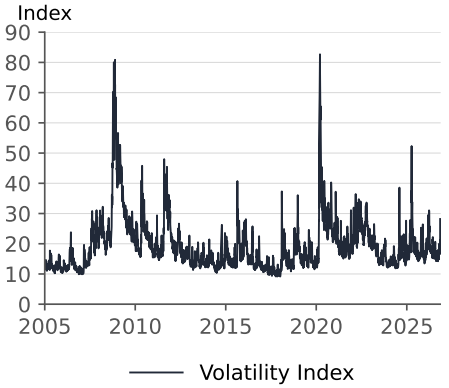


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US Stock Volatility Index (VIX)



Measuring Volatility, Variance and Standard Deviation

- **Variance**, the average squared deviation from the average return, in four steps,
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- **Standard deviation**, the square root of the variance,

$$\text{Standard deviation} = \sqrt{\text{Variance}}$$

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Year	Return
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6	20%

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Average	10%	Variance (average of squared deviations)	0.0100

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Average	10%	Variance (average of squared deviations)	0.0100
		Standard deviation (square root)	10%

Standard Deviation of Treasury Bills vs. Bonds vs. Stocks

Portfolio	Average annual return 1928–2025	Standard deviation 1928–2025
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Tesla	201.4
Amazon	186.9
Apple	65.1
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- This is the effect of diversification

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- Each has a good year or a bad year next year, depending on its own events (say, its product succeeds or flops)
- The two companies compete, so one tends to have a good year when the other has a bad year
- The four scenarios are not equally likely,

Scenario	Probability	Stock A (%)	Stock B (%)
A good, B good	20%	+25	+20
A good, B bad	30%	+25	−10
A bad, B good	30%	−5	+20
A bad, B bad	20%	−5	−10

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- As investors, we love expected return but hate volatility (standard deviation)
- So we want a high expected return for a low standard deviation
- Then how should we invest?

Example, Portfolio Diversification

- What are the expected return and the volatility if we invest in one stock?

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Stock A	Return
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$E[R_A]$	10%

Stock B	Return
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$E[R_B]$	5%

Example, Portfolio Diversification

- What are the expected return and the volatility if we invest in one stock?

Stock A	Return	Deviation from $E[R_A]$	
Good year	25%	15%	
Bad year	-5%	-15%	
$E[R_A]$	10%		

Stock B	Return	Deviation from $E[R_B]$	
Good year	20%	15%	
Bad year	-10%	-15%	
$E[R_B]$	5%		

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Stock A	Return	Deviation from $E[R_A]$	Squared deviation
Good year	25%	15%	0.0225
Bad year	-5%	-15%	0.0225
$E[R_A]$	10%		

Stock B	Return	Deviation from $E[R_B]$	Squared deviation
Good year	20%	15%	0.0225
Bad year	-10%	-15%	0.0225
$E[R_B]$	5%		

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Stock A	Return	Deviation from $E[R_A]$	Squared deviation
Good year	25%	15%	0.0225
Bad year	-5%	-15%	0.0225
$E[R_A]$	10%	Variance	0.0225

Stock B	Return	Deviation from $E[R_B]$	Squared deviation
Good year	20%	15%	0.0225
Bad year	-10%	-15%	0.0225
$E[R_B]$	5%	Variance	0.0225

Example, Portfolio Diversification

- What are the expected return and the volatility if we invest in one stock?

Stock A	Return	Deviation from $E[R_A]$	Squared deviation
Good year	25%	15%	0.0225
Bad year	−5%	−15%	0.0225
$E[R_A]$	10%	Variance	0.0225
		Std. dev.	15.0%

Stock B	Return	Deviation from $E[R_B]$	Squared deviation
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Bad year	−10%	−15%	0.0225
$E[R_B]$	5%	Variance	0.0225
		Std. dev.	15.0%

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- $E[R]$ / std. dev., stock A $10/15 = 0.67$, stock B $5/15 = 0.33$

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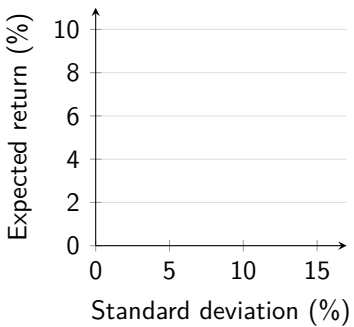
Stock B	Return	Deviation from $E[R_B]$	Squared deviation
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$E[R_B]$	5%	Variance	0.0225
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- $E[R]$ / std. dev., stock A $10/15 = 0.67$, stock B $5/15 = 0.33$
- Stock A looks better as a stand-alone investment on this metric

Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

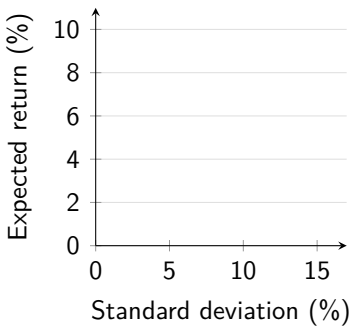
Portfolio	
Stock A weight	
Stock B weight	
A good, B good	
A good, B bad	
A bad, B good	
A bad, B bad	
$E[R]$	
Std. dev.	
$E[R]$ / std. dev.	



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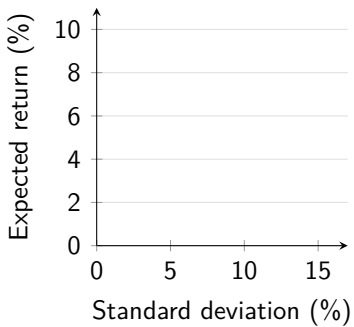
Portfolio	1
Stock A weight	100%
Stock B weight	
A good, B good	
A good, B bad	
A bad, B good	
A bad, B bad	
$E[R]$	
Std. dev.	
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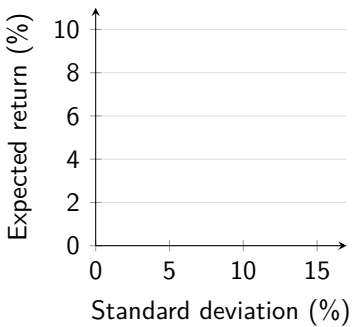
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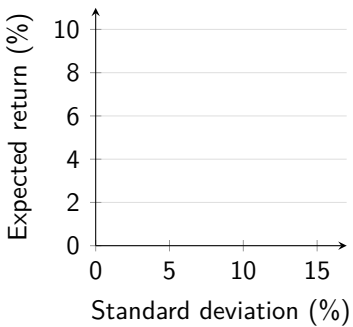
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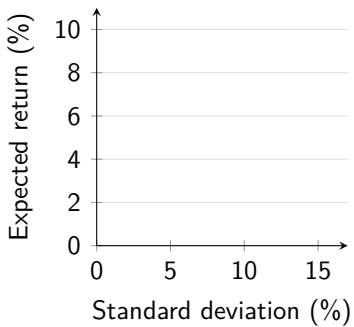
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Stock B weight	0%
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A bad, B good	
A bad, B bad	
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Std. dev.	
$E[R]$ / std. dev.	



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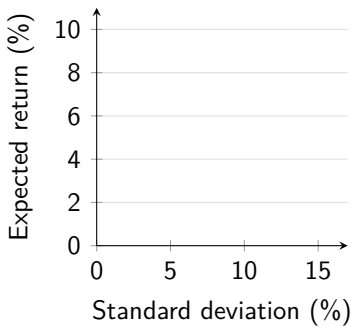
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A bad, B bad	
$E[R]$	
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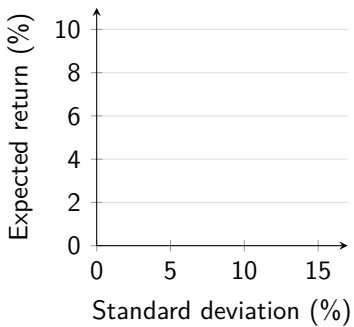
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$E[R]$	
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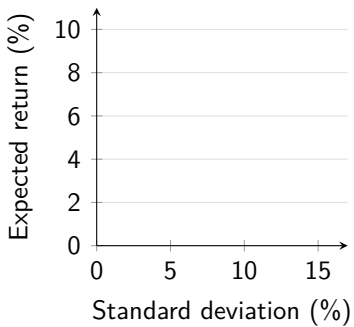
Portfolio	1
Stock A weight	100%
Stock B weight	0%
A good, B good	25%
A good, B bad	25%
A bad, B good	−5%
A bad, B bad	−5%
$E[R]$	10.0%
Std. dev.	
$E[R]$ / std. dev.	



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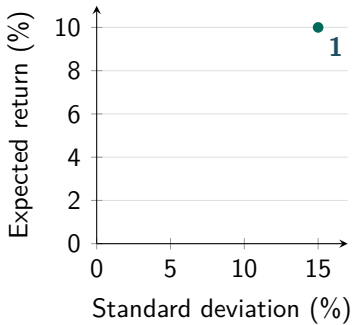
Portfolio	1
Stock A weight	100%
Stock B weight	0%
A good, B good	25%
A good, B bad	25%
A bad, B good	−5%
A bad, B bad	−5%
$E[R]$	10.0%
Std. dev.	15.0%
$E[R]$ / std. dev.	



Example, Portfolio Diversification

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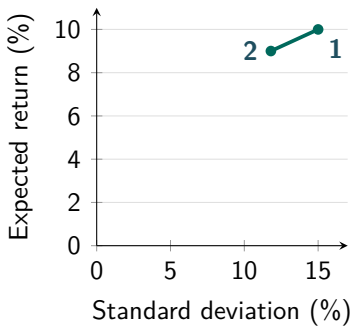
Portfolio	1
Stock A weight	100%
Stock B weight	0%
A good, B good	25%
A good, B bad	25%
A bad, B good	−5%
A bad, B bad	−5%
$E[R]$	10.0%
Std. dev.	15.0%
$E[R]$ / std. dev.	0.67



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

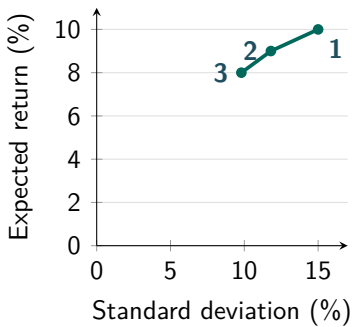
Portfolio	1	2
Stock A weight	100%	80%
Stock B weight	0%	20%
A good, B good	25%	24%
A good, B bad	25%	18%
A bad, B good	−5%	0%
A bad, B bad	−5%	−6%
$E[R]$	10.0%	9.0%
Std. dev.	15.0%	11.8%
$E[R]$ / std. dev.	0.67	0.76



Example, Portfolio Diversification

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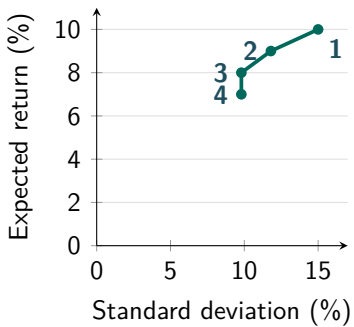
Portfolio	1	2	3
Stock A weight	100%	80%	60%
Stock B weight	0%	20%	40%
A good, B good	25%	24%	23%
A good, B bad	25%	18%	11%
A bad, B good	−5%	0%	5%
A bad, B bad	−5%	−6%	−7%
$E[R]$	10.0%	9.0%	8.0%
Std. dev.	15.0%	11.8%	9.8%
$E[R]$ / std. dev.	0.67	0.76	0.82



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

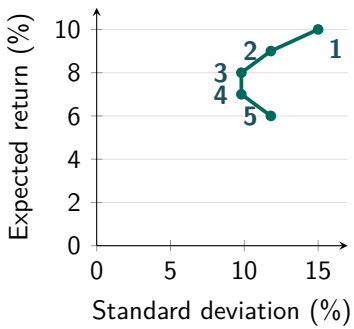
Portfolio	1	2	3	4
Stock A weight	100%	80%	60%	40%
Stock B weight	0%	20%	40%	60%
A good, B good	25%	24%	23%	22%
A good, B bad	25%	18%	11%	4%
A bad, B good	−5%	0%	5%	10%
A bad, B bad	−5%	−6%	−7%	−8%
$E[R]$	10.0%	9.0%	8.0%	7.0%
Std. dev.	15.0%	11.8%	9.8%	9.8%
$E[R]$ / std. dev.	0.67	0.76	0.82	0.72



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

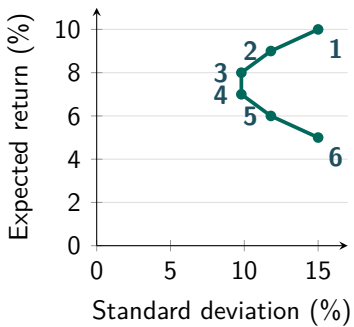
Portfolio	1	2	3	4	5
Stock A weight	100%	80%	60%	40%	20%
Stock B weight	0%	20%	40%	60%	80%
A good, B good	25%	24%	23%	22%	21%
A good, B bad	25%	18%	11%	4%	−3%
A bad, B good	−5%	0%	5%	10%	15%
A bad, B bad	−5%	−6%	−7%	−8%	−9%
$E[R]$	10.0%	9.0%	8.0%	7.0%	6.0%
Std. dev.	15.0%	11.8%	9.8%	9.8%	11.8%
$E[R]$ / std. dev.	0.67	0.76	0.82	0.72	0.51



Example, Portfolio Diversification

- What if we mixed the two stocks with different weights?

Portfolio	1	2	3	4	5	6
Stock A weight	100%	80%	60%	40%	20%	0%
Stock B weight	0%	20%	40%	60%	80%	100%
A good, B good	25%	24%	23%	22%	21%	20%
A good, B bad	25%	18%	11%	4%	−3%	−10%
A bad, B good	−5%	0%	5%	10%	15%	20%
A bad, B bad	−5%	−6%	−7%	−8%	−9%	−10%
$E[R]$	10.0%	9.0%	8.0%	7.0%	6.0%	5.0%
Std. dev.	15.0%	11.8%	9.8%	9.8%	11.8%	15.0%
$E[R]$ / std. dev.	0.67	0.76	0.82	0.72	0.51	0.33



Portfolio Diversification

- What is the standard deviation of the portfolio?

$$\text{StdDev}[R_P] \neq w_A \,\text{StdDev}[R_A] + w_B \,\text{StdDev}[R_B]$$

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- **Diversification**, spreading the money in a portfolio across many securities rather than one

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- **Diversification**, spreading the money in a portfolio across many securities rather than one
- Diversification typically makes the standard deviation of a portfolio lower than the weighted average of the standard deviations of the stocks in it,

$$\textbf{StdDev}[R_P] < w_A \,\text{StdDev}[R_A] + w_B \,\text{StdDev}[R_B]$$

Simulation with AI

- An economy with N companies, each producing a product

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Simulation with AI

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- Two kinds of risk,
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  - A company's product fails (30% chance each year), its stock return falls by 40 points
  - Something bad hits the whole economy, like a pandemic (10% chance each year), every stock return falls by 30 points

Simulating an Investor's Return

- Simulate many years of this economy, the AI writes the code and plots

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- Now the investor splits the money equally across 2 companies, then 5, 10, 30, 100. Does the standard deviation fall? To zero?

What Is the End Game of Portfolio Diversification?

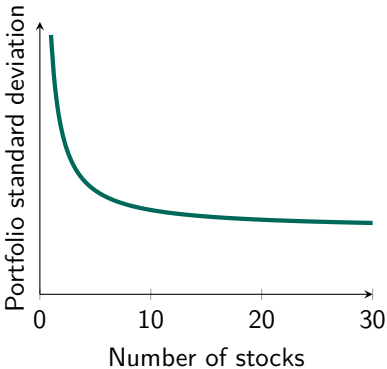
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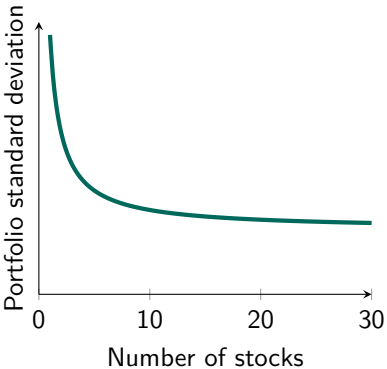
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- The more stocks, the lower it goes



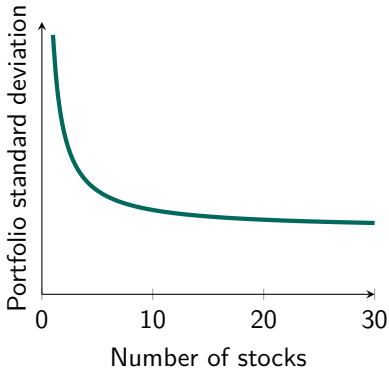
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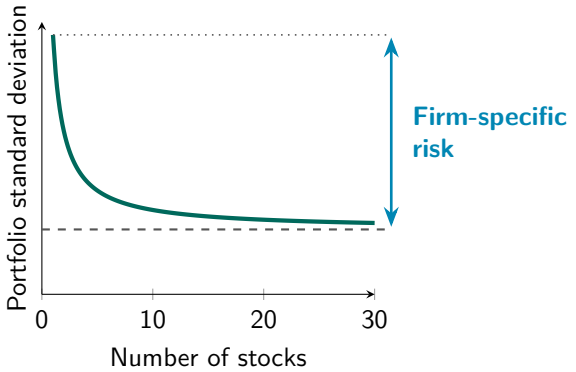
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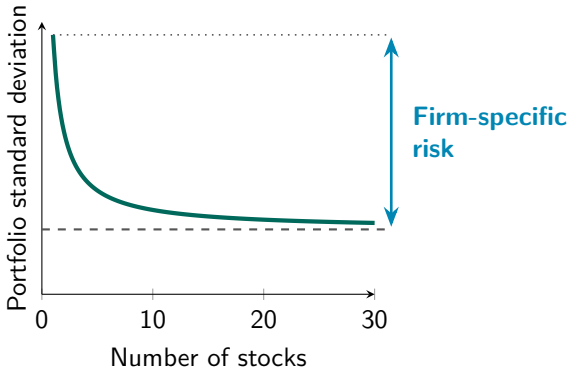
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- Such events average out across many stocks
- Diversification removes firm-specific risk



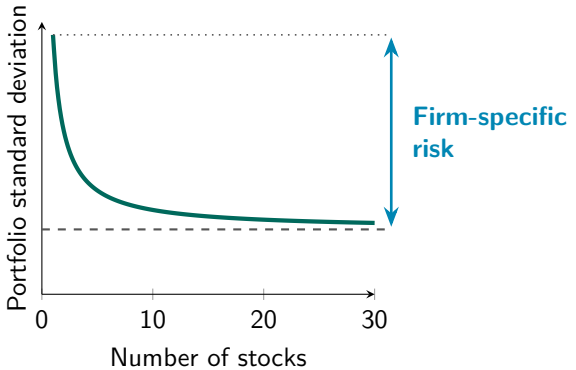
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- Yet there is a limit, the curve never reaches zero



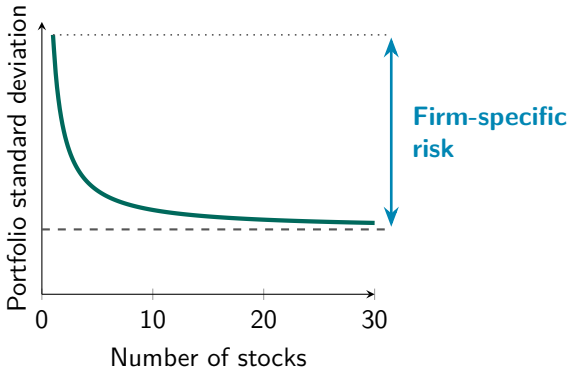
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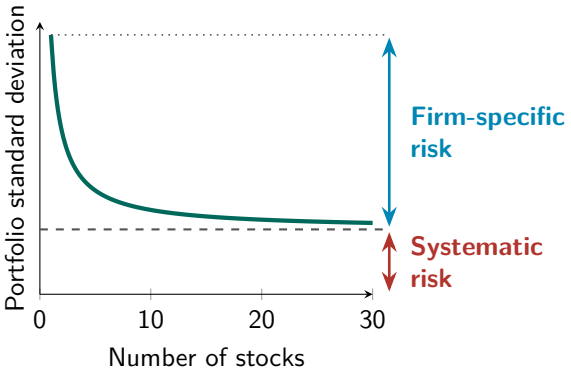
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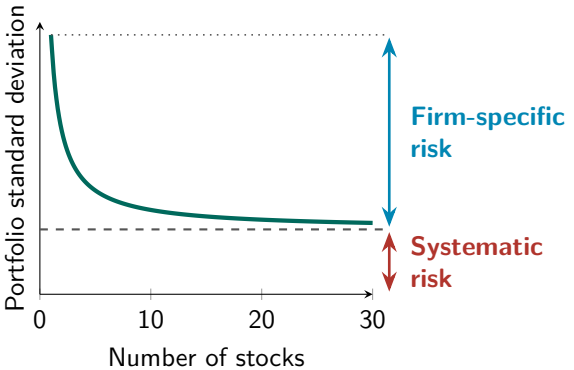
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The end game, a diversified portfolio carries systematic risk only

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- **Volatility**, how much returns vary, the **variance** or **standard deviation**

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- Stocks vary far more than bonds and Treasury bills, individual stocks even more
- **Diversification**, a portfolio's standard deviation is below its stocks' average
- Diversification removes firm-specific risk, not systematic risk

Module 8, Part 3

Beta

From Standard Deviation to Beta

- **Part 2**'s lesson, a diversified investor is left with systematic risk only

From Standard Deviation to Beta

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Beta

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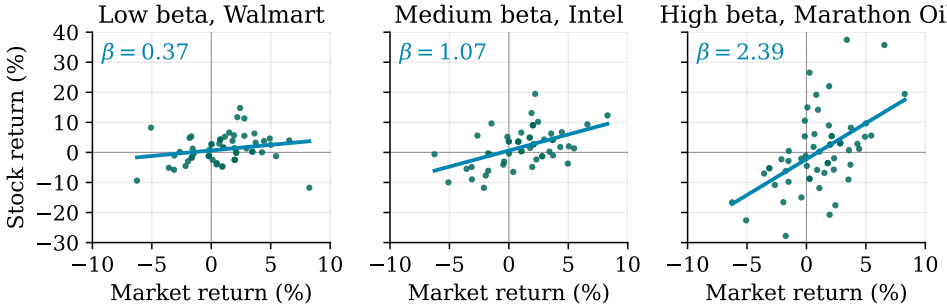
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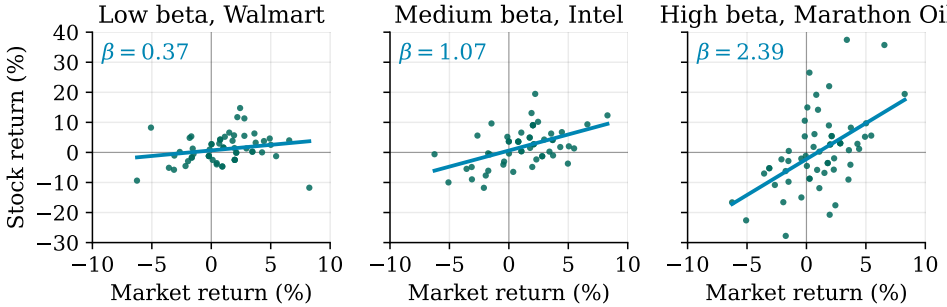
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Example, Low vs. Medium vs. High Beta Stocks

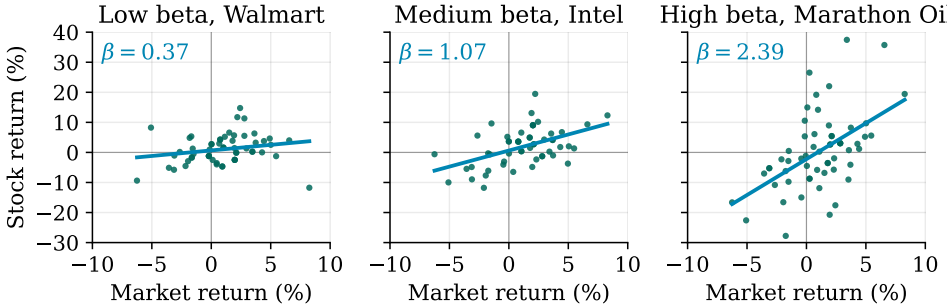


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  - Its return moves one for one with the market's, because it is the market

A Primer on Regression Analysis

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- **Regression analysis**, a statistical tool for describing the "best-fit" linear relation between two variables, X (the "explanatory" variable) and Y (the "dependent" variable)

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- The best-fit line keeps the residuals e_i as small as possible overall

Example, Height and Weight

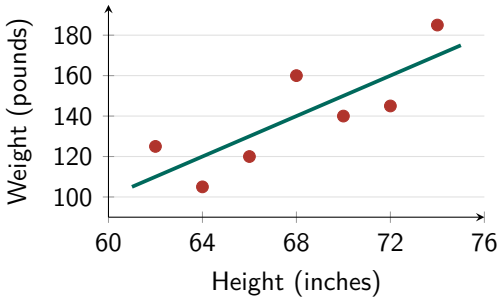
- Seven people's heights and weights, what is the best-fit line between weight (Y) and height (X)?

Person	Height (inches)	Weight (pounds)
1	62	125
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- Solution, the best-fit line is Weight $= -200 + 5 \times$ Height, so $\alpha = -200$ and $\beta = 5$

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- The coefficient β from this regression is the estimate of the stock's beta

Example, Measuring Beta

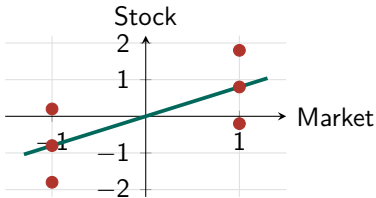
- A company's stock has the following excess returns, alongside the market portfolio's (both in %). What is its beta?

Month	Market excess return	Stock excess return
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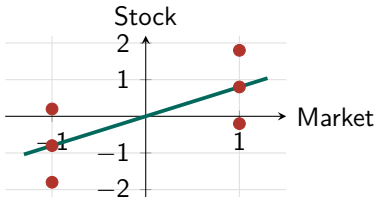


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- The slope is the estimate of the stock's beta, here $\beta = 0.8$

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Module 8, Part 4

The CAPM

CAPM, the Relation between Risk (Beta) and Expected Return

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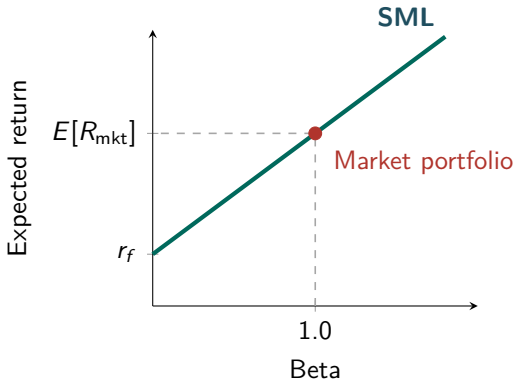
$$E[R_i] = r_f + \beta_i \times (E[R_{\text{mkt}}] - r_f)$$

  - $E[R_{\text{mkt}}] - r_f$, the **market risk premium**, the expected return on the market portfolio over the risk-free rate

CAPM, the Security Market Line

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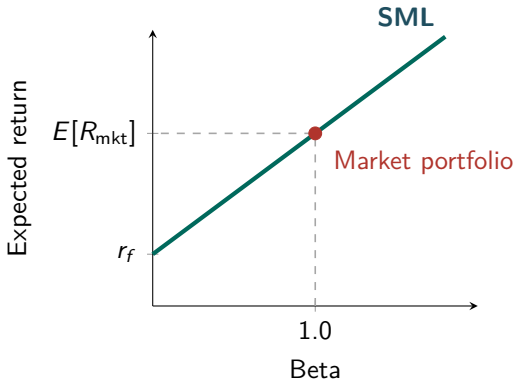


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- A straight line, it starts at r_f for $\beta = 0$ and passes through the market portfolio, $\beta = 1$ and $E[R_{\text{mkt}}]$



CAPM, Expected Returns for Selected Companies

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Company	Business	Beta
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- It earned him the Nobel Prize in economics in 1990



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c. Investors demand higher expected rates of return from stocks with returns that are highly exposed to macroeconomic changes

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c. True. The asset's beta is a function of its sensitivity to macroeconomic risks, among other factors

Recap of Part 4

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Module 8, Part 5

From Expected Return to Discount Rate

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- Is this return the project's discount rate r in capital budgeting?

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Discount rate of a project = Expected return on stocks with the same beta

Capital Budgeting with the CAPM, Three Steps

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Capital Budgeting with the CAPM, Three Steps

1. Estimate the beta of the project
2. Plug it into the CAPM to get the discount rate,
$$r = r_f + \beta_{\text{project}} \times (E[R_{\text{mkt}}] - r_f)$$
3. Discount the project's cash flows at r as in **Module 7**, accept if NPV > 0 (equivalently, if IRR $> r$)

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- Same cash flows, opposite answers. The discount rate comes from the project's beta, and accepting only the projects with NPV > 0 at this rate makes the shareholders better off

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- The company is a portfolio of its two businesses, with weights 80% and 20%, so its beta is the weighted average (**Part 3**),

$$\beta_{\text{company}} = 0.8 \times \beta_{\text{soup}} + 0.2 \times \beta_{\text{oil}} = 0.8 \times 0.5 + 0.2 \times 2.0 = 0.8$$

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- That is exactly the weighted average of each business's own CAPM rate,

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- So discount the oil project at 17%, its own rate, not at the company's 6.5%

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- A company can set two or more discount rates for different types of projects

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- Use the project's own beta (the twin method), not the company's

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- **Module 9**, the discount rate becomes the weighted average cost of capital (WACC), a mix of what lenders and shareholders require

Thank you!

Please let me know if you have any questions.